

HDOC-22

Question:

Write a menu-driven C program that accepts two positive integers, m and n , where $m \leq n$. Display the following menu and use a switch statement to perform the operation selected by the user:

- 1. Print the multiplication tables of m and n side by side, using multipliers from 1 to 10.*
- 2. Display a multiplication grid containing the products of integers from 1 to m and integers from 1 to n .*
- 3. For each integer from m to n , print that integer followed by the next five consecutive integers.*
- 4. Exit the program.*

Continue displaying the menu until the user selects the Exit option. Display an appropriate error message if the user enters an invalid menu choice.

Task1: Algorithm:

1. Algorithm

Step 1: Start.

Step 2: Declare integer variables m , n , choice, i , and j .

Step 3: Input two positive integers m and n .

Step 4: Check the input:

- If $m \leq 0$ or $n \leq 0$ or $m > n$, display "Invalid input" and stop.
- Otherwise, continue.

Step 5: Display the menu:

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Step 6: Input choice.

Step 7: Use a switch statement to perform the selected operation.

Case 1: Multiplication tables

1. Set $i = 1$.
2. Repeat while $i \leq 10$.

3. Calculate and display:

- $m \times i$
- $n \times i$

4. Increment i.

5. Return to the menu.

Case 2: Multiplication grid

1. Display column numbers from 1 to n.

2. Set $i = 1$.

3. Repeat while $i \leq m$.

4. For each i, set $j = 1$.

5. Repeat while $j \leq n$.

6. Calculate and display $i * j$.

7. Increment j.

8. After completing one row, increment i.

9. Return to the menu.

Case 3: Consecutive integers

1. Set $i = m$.

2. Repeat while $i \leq n$.

3. Display i.

4. Display the six integers from i through $i + 5$.

5. Increment i.

6. Return to the menu.

Case 4: Exit

Display "Exiting program." and terminate the loop.

Default:

Display:

Invalid choice. Please enter a choice from 1 to 4.

Return to the menu.

Step 8: Continue displaying the menu until the user selects 4.

Step 9: Stop.

Task2: Test case table:

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	m=3, n=5, Choice 1	Tables of 3 and 5 from $\times 1$ to $\times 10$	Tables of 3 and 5 from $\times 1$ to $\times 10$	Pass
2	m=3, n=5, Choice 2	3×5 multiplication grid	Correct 3×5 grid displayed	Pass
3	m=3, n=5, Choice 3	3: 3 4 5 6 7 8, 4: 4 5 6 7 8 9, 5: 5 6 7 8 9 10	Same output obtained	Pass
4	m=3, n=5, Choice 4	Program exits	Exiting program.	Pass
5	m=3, n=5, Choice 5	Invalid-choice error	Invalid-choice error displayed	Pass
6	m=1, n=1, Choice 1	Table of 1 displayed twice	Correct output obtained	Pass
7	m=1, n=1, Choice 2	Grid containing $1 \times 1 = 1$	Correct output obtained	Pass
8	m=1, n=1, Choice 3	1: 1 2 3 4 5 6	Correct output obtained	Pass
9	m=5, n=3	Invalid because $m > n$	Invalid-input message	Pass
10	m=0, n=5	Invalid because m is not positive	Invalid-input message	Pass
11	m=-2, n=5	Invalid because m is negative	Invalid-input message	Pass

Task3: C program:

```
#include <stdio.h>
int main()
{
    int m, n, choice, i, j;
    printf("Enter two positive integers m and n (m <= n): ");
    scanf("%d %d", &m, &n);

    /* Check whether m and n are valid */
    if (m <= 0 || n <= 0 || m > n)
    {
        printf("Invalid input. Enter positive integers such that m <= n.\n");
        return 0;
    }
    do
    {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n", m,
n, m, n);
                for (i = 1; i <= 10; i++)
                {
                    printf("%d x %d = %-5d    %d x %d = %d\n",
                        m, i, m * i,
                        n, i, n * i);
                }
                break;
            case 2:
                printf("\nMultiplication Grid (m=%d, n=%d)\n\n",
                    m, n);
                /* Column headings */
                printf("    ");
                for (j = 1; j <= n; j++)
                {
                    printf("%4d", j);
```

```

    }
    printf("\n  ");
    for (j = 1; j <= n; j++)
    {
        printf("----");
    }
    printf("\n");
    /* Multiplication grid */
    for (i = 1; i <= m; i++)
    {
        printf("%2d |", i);
        for (j = 1; j <= n; j++)
        {
            printf("%4d", i * j);
        }
        printf("\n");
    }
    break;
case 3:
    printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n",
        m, n, m, n);
    for (i = m; i <= n; i++)
    {
        printf("%d:", i);
        for (j = i; j <= i + 5; j++)
        {
            printf(" %d", j);
        }
        printf("\n");
    }
    break;
case 4:
    printf("Exiting program.\n");
    break;
default:
    printf("Invalid choice. Please enter a choice from 1 to 4.\n");
}
} while (choice != 4);
return 0;
}

```

Task4: Output:

```
(kali㉿kali)-[~/hdoc22.c]
$ ./a.out
Enter two positive integers m and n (m≤n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 3 and 5(m=3, n=5)
3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

Multiplication Grid (m=3, n=5)

  1  2  3  4  5
  --
1  1  2  3  4  5
2  2  4  6  8 10
3  3  6  9 12 15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3

Consecutive Integers from 3 to 5(m=3, n=5)
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4
exiting program.
```

```

(kali㉿kali)-[~/hdoc22.c]
$ ./a.out
Enter two positive integers m and n (m≤n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 5
Invalid choice. Please enter a choice from 1 to 4.

```

```

(kali㉿kali)-[~/hdoc22.c]
$ ./a.out
Enter two positive integers m and n (m≤n): 1 1

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 1 and 1(m=1, n=1)
1 x 1 = 1      1 x 1 = 1
1 x 2 = 2      1 x 2 = 2
1 x 3 = 3      1 x 3 = 3
1 x 4 = 4      1 x 4 = 4
1 x 5 = 5      1 x 5 = 5
1 x 6 = 6      1 x 6 = 6
1 x 7 = 7      1 x 7 = 7
1 x 8 = 8      1 x 8 = 8
1 x 9 = 9      1 x 9 = 9
1 x 10 = 10    1 x 10 = 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

Multiplication Grid (m=1, n=1)

  1
  —
1  1

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3

Consecutive Integers from 1 to 1(m=1, n=1)
1: 1 2 3 4 5 6

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4
exiting program.

```

```
(kali㉿kali)-[~/hdoc22.c]  
$ ./a.out  
Enter two positive integers m and n ( $m \leq n$ ): 5 3  
Invalid input. Enter positive integers such that  $m \leq n$ .
```

```
(kali㉿kali)-[~/hdoc22.c]  
$ ./a.out  
Enter two positive integers m and n ( $m \leq n$ ): 0 5  
Invalid input. Enter positive integers such that  $m \leq n$ .
```

```
(kali㉿kali)-[~/hdoc22.c]  
$ ./a.out  
Enter two positive integers m and n ( $m \leq n$ ): -2 5  
Invalid input. Enter positive integers such that  $m \leq n$ .
```